

SUMMARY

Data Analyst with 3+ years of experience in fraud analytics, regulatory compliance, and large-scale data processing. Improved fraud detection accuracy by 18% and reduced false positives by 22% through data-driven analysis. Built an AI-driven FDA compliance system to extract and structure unstructured inspection data for regulatory intelligence and risk assessment. Proficient in Python, SQL, ETL pipelines, and data visualization, with a strong ability to translate complex data into actionable insights through cross-functional collaboration.

TECHNICAL SKILLS

Programming: Python, SQL, MATLAB, R, Bash

Big Data & Distributed Systems: Apache Spark, PySpark, Spark SQL, Hive, Hadoop, Databricks

ML & LLM Data: Scikit-learn, PyTorch, LLM evaluation, Prompt Engineering

Visualization: Tableau, Power BI, Matplotlib, Seaborn, Plotly

Cloud: AWS, GCP, Azure Data Studio

ETL & Pipelines: Data Wrangling, PySpark ETL, API Integration, Data Curation, Privacy-preserving ETL

Databases & Warehouses: Snowflake, MySQL, PostgreSQL

Tools: Git, VS Code, Jupyter, Docker, Agile/Scrum

Other Technical Skills: REST APIs, JSON, Regex, Web Scraping, Data Pipeline Development, Statistical Modeling

PROFESSIONAL EXPERIENCE

Data Analytics Intern

Aug 2025 - Dec 2025

Precise Software Solutions, Inc

Fairfax, VA

- Integrated structured metadata and unstructured PDF inspection reports using Python to build the foundation of a compliance system, which organized inspection tracking and enabled regulatory intelligence extraction.
- Developed a preprocessing pipeline using PyPDF2 and regex in Python to extract, clean, and standardize narrative observations from long-form FDA Form 483 PDFs, significantly reducing manual data-entry effort and improving data consistency.
- Cross-validated extracted observations with key FDA metadata to ensure high accuracy and alignment with official records.
- Implemented an LLM-based classification model to assign inspection status (OAI, VAI, NAI), violation severity, compliance program mapping, and recommended follow-up actions, accelerating report analysis and supporting automated decision making.
- Structured outputs in strict JSON format, providing machine-readable data for dashboards, analytics, chatbot integrations, and automated regulatory intelligence to support risk assessment and compliance monitoring.

Research Assistant – Data Analytics & Machine Learning

Jan 2025 - July 2025

George Mason University

Fairfax, VA

- Processed large-scale transactional datasets on Databricks Spark clusters, enabling fast and scalable analytics.
- Designed and implemented distributed ETL workflows with PySpark and Snowflake for data ingestion, cleaning, and transformation, reducing processing time and enabling timely reporting.
- Generated optimized association rules using FP-Growth, improving recommendation quality and model efficiency.
- Developed parameterized notebooks to automate the pipeline, ensuring repeatability and scalability across experiments.
- Provided insights on purchasing patterns using Seaborn visualizations and Snowflake queries, supporting data-driven marketing and product placement strategies.

Research Assistant – Predictive Analytics & Data Modeling

Aug 2024 - Dec 2024

George Mason University

Fairfax, VA

- Processed a dataset of 59,500 rows and 17 features using Spark SQL and privacy-preserving ETL scripts, automating cleaning and feature transformation, which streamlined data preparation for downstream modeling.
- Developed an end-to-end predictive-analytics preprocessing pipeline in Spark SQL on Hadoop that performed imputation, encoding, normalization, and scaling, enabling consistent dataset preparation for ML models and accelerating model training.
- Implemented a Random Forest model with scikit-learn and optimized hyperparameters via GridSearchCV, which improved prediction accuracy for bridge maintenance planning.
- Produced tabular reports that presented predictive insights, enabling bridge engineers to prioritize maintenance activities.
- Enhanced model interpretability and decision-making by analyzing feature importance and performance metrics.

Data Analyst

May 2022 - Dec 2023

Claro Software Solutions Pvt Ltd

Hyderabad, India

- Conducted in-depth fraud detection analysis on over 250,000 transaction records, identifying behavioral patterns and implementing data-driven algorithms that improved detection accuracy by 18%.
- Collaborated with a cross-functional team to enhance existing fraud detection models, reducing false positives by 22% and improving precision in identifying legitimate transactions.
- Presented analytical insights and findings to senior management using interactive Tableau dashboards, which led to the adoption of updated fraud-detection workflow guidelines.
- Utilized Python, SQL, and machine-learning techniques on GCP BigQuery to analyze large datasets, delivering actionable insights that accelerated the investigation of suspicious transactions.
- Developed interactive dashboards and visual reports to monitor fraud trends and enabling teams to make informed decisions.

EDUCATION

George Mason University

Fairfax, VA

Master of Science in Data Analytics Engineering | GPA: 3.90

Jan 2024 – Dec 2025